

Deriving Unsupervised SIMCSE From First Principles

Goal: Given a Batch of unlabeled sentences, learn an encoder whose representations identify two stochastic views of same sentence while distinguishing representations of other sentences

Step 1:- $x_i = i^{\text{th}}$ sentence

Eg:- $x_i = \text{"The human is running"}$

How can we create positive pairs without having another sentence.
→ Use same sentence twice

So, $x_i^1 = x_i$, $x_i^2 = x_i$

But if I pass information / exact same input through same deterministic network, I will get same representation

- So, to make 2 represⁿ slightly different without changing sentence
→ Dropout

$$h_i = f_\theta(x_i, z_i), \quad h_i^+ = f_\theta(x_i, z_i')$$

f_θ = Best with parameters θ

z_i = First Dropout mask

z_i' = Second ~

And $z_i \neq z_i'$, So, $h_i \neq h_i^+$

As Dropout are independently sampled,

So,

(h_i, h_i^+) is positive pair
as represⁿ corresponds to x_i

Dropout was needed as gives 2 slightly diff^t represⁿ of same sentence
allowing model to learn invariance to small stochastic perturbations

Suppose Batch $B = \{x_1, x_2, \dots, x_n\}$

Each sentence goes through Encoder twice

$$x_1 \rightarrow h_1, h_1^+$$

$$x_2 \rightarrow h_2, h_2^+$$

$\vdots$

$$x_n \rightarrow h_n, h_n^+$$

Now, for h_i , correct repⁿ match
is h_i^+

So, while (h_i, h_i^+) is the pair
 $h_j^+, j \neq i$, comes from diff
sentences, so they become negative
pairs, in-batch negatives

Suppose $N=4$ for h_1
Candidates = $[h_1^+, h_2^+, h_3^+, h_4^+]$
Answer = h_1^+

So, Sim(S) is asking classification
problem,

For h_i , correct class is $y_i = i$
 So loss can be implemented
 using ordinary cross entropy

Which candidate is similar?

→ Cosine Similarity

$$S_{ij} = \text{sim}(h_i, h_j^+) = \frac{h_i^T h_j^+}{\|h_i\| \|h_j^+\|}$$

3 pairs, same semantic meaning
 $\Rightarrow$ large similarity

$$S = \begin{bmatrix} S_{11} & S_{12} & \dots & S_{14} \\ S_{21} & \dots & \dots & S_{24} \\ S_{31} & \dots & \dots & S_{34} \\ S_{41} & \dots & \dots & S_{44} \end{bmatrix}$$

Interpretation:-

anchors h_i	Candidates			
	h_1^+	h_2^+	h_3^+	h_4^+
h_1	✓	x	x	x
h_2	x	✓	x	x
h_3	x	x	✓	x
h_4	x	x	x	✓

So objective is make S_{ij} large
and S_{ij} small ($i \neq j$)

Turning similarity scores into
probabilities (softmax)

$$P(j|i) = \frac{\exp(S_{ij})}{\sum_{k=1}^N \exp(S_{ik})}$$

Adding Temperature

$$\Rightarrow P(j|i) = \frac{\exp(S_{ij} / T)}{\sum_{k=1}^N \exp(S_{ik} / T)}$$

P_{ij} needs to be maximised

$$P(i|i) = \frac{\exp(S_{ii} / T)}{\sum_{k=1}^N \exp(S_{ik} / T)}$$

So, we can minimize negative log
probability

$$L_i = -\log P(i|i)$$

$$l_i = -\log \frac{\exp(s_{ii} / \tau)}{\sum_{k=1}^N \exp(s_{ik} / \tau)}$$

as, $s_{ij} = \text{sim}(h_i, h_j^+)$

$$l_i = -\log \frac{\exp(\text{sim}(h_i, h_i^+) / \tau)}{\sum_{j=1}^N \exp(\text{sim}(h_i, h_j^+) / \tau)}$$

Now avg across batch
 $l_1, l_2, \dots, l_n$

Batch loss $\Rightarrow L = \frac{1}{N} \sum_{i=1}^N l_i$

$$\Rightarrow L = \frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(h_i, h_i^+) / \tau)}{\sum_{j=1}^N \exp(\text{sim}(h_i, h_j^+) / \tau)}$$

Now, Similarities = [0.9, 0.2, 0.1]
 Positive = candidate 0
 Temp = 0.5

Probabilities $\Rightarrow$ Candidate 0: 0.6904
1: 0.1702
2: 0.1394
Positive Probab.

$$\text{Loss} = 0.3705 \text{ (for positive)}$$

This is good, as Candidate 0 was supposed to be 75% and highest and loss was also less, as it was -ve log Probab, as it was correct represent

If Similarities = [0.7, 0.99, 0.01]
Positive = B Candidate 1
Temp = 0.5

Probabilities $\Rightarrow$ Candidate 0: 0.3292
1: 0.5880
2: 0.0828
Positive

$$\text{Loss} = 0.5311$$

If Candidate = 0
Loss = 1.1111

If Temp = 1.0, Loss Sim = [0.9, 0.2, 0.1]
Prob. = [0.5139, 0.2552, 0.2309]
 ↓
 true, loss = 0.6657

" " = 0.5, Prob = [0.6904, 0.1702, 0.1302]
Loss = 0.3705

" " = 0.1, Prob = [0.9988, 0.0009, 0.0003]
Loss = 0.0012

As temp decreases, diff. b/w Sim. ↑

If trues stay fixed, the probab. ↑

If -ve & ↑ similar to anchor
Loss ↑

If true has very high sim, negative
have very low, model should
assign almost all probab. to positive
So, loss very low